

The Free Rational Calogero–Moser Flow as a Hermitian Pencil: Global Completeness, Trace Integrals, and a Spectral Atlas

Route-A structural certificate C196

Abstract

For every particle number $N \geq 2$, repulsive coupling $g > 0$, ordered real initial positions, and real momenta, we recover the rational Calogero–Moser trajectory as the ordered spectrum of one Hermitian pencil $Q_0 + tL_0$. A rank-one commutator proves that this pencil is simple for every real physical time. Eigenvalue perturbation then gives the Hamilton equations with the exact force coefficient $2g^2$, hence collision-free global completeness, while unitary transport of L_0 gives every trace integral and $\text{Tr } L^2 = 2H$. Diagonalizing L_0 also gives a global spectral atlas: ordered velocities and arbitrary real intercepts reconstruct one and only one ordered phase point. These are classical Calogero–Moser/Moser structures assembled with explicit convention and scope locks, not a new priority claim.

Keywords: Calogero–Moser system; Hermitian pencil; Lax pair; complete flow; isospectral integrals.

中文摘要

对任意粒子数 $N \geq 2$ 、排斥耦合 $g > 0$ 、严格有序的实初始位置和实动量，本文把有理 Calogero–Moser 轨道写成单个厄米矩阵束 $Q_0 + tL_0$ 的有序谱。秩一交换子证明该谱在全部物理时间内保持单重；特征值微扰给出系数恰为 $2g^2$ 的哈密顿方程，从而同时得到无碰撞与全局完备性。 L_0 的西共轭还给出全部迹积分以及 $\text{Tr } L^2 = 2H$ 。对角化 L_0 还得到一个全局谱坐标图：有序速度与任意实截距唯一重建有序相点。这些结论属于经典理论的约定锁定与整体闭合，并不主张经典定理优先权。
关键词：Calogero–Moser 系统；厄米矩阵束；Lax 对；全局完备性。

1 Frozen Hamiltonian and matrix convention

On the chamber $q_1 < \dots < q_N$, set

$$H(q, p) = \frac{1}{2} \sum_{j=1}^N p_j^2 + \sum_{1 \leq j < k \leq N} \frac{g^2}{(q_j - q_k)^2}, \quad g > 0. \quad (1)$$

With $Q_0 = \text{diag}(q_1, \dots, q_N)$, define the Hermitian matrix

$$(L_0)_{jk} = p_j \delta_{jk} + \frac{ig(1 - \delta_{jk})}{q_j - q_k}, \quad X(t) = Q_0 + tL_0. \quad (2)$$

The sign, unordered-pair normalization in (1), and physical clock t are part of the claim. Direct multiplication gives

$$[Q_0, L_0] = ig(J - I), \quad J = ee^*, \quad e = (1, \dots, 1)^T, \quad (3)$$

and

$$\text{Tr } L_0^2 = \sum_j p_j^2 + 2 \sum_{j < k} \frac{g^2}{(q_j - q_k)^2} = 2H. \quad (4)$$

Calogero’s inverse-square model and Moser’s isospectral solution retain classical ownership [1, 2].

2 A rank-one proof of simplicity

For all real t , $[X(t), L_0] = [Q_0, L_0] = ig(J - I)$. Suppose an eigenspace E of $X(t)$ had dimension at least two, and let P_E be its orthogonal projector. Because $X(t)|_E$ is scalar,

$$P_E[X(t), L_0]P_E = 0.$$

Equation (3) would force $P_E J P_E = P_E$. The left side has rank at most one, whereas the right side has rank at least two, a contradiction. Thus the increasingly ordered eigenvalues

$$x_1(t) < \dots < x_N(t)$$

of $X(t)$ exist without a crossing for every $t \in \mathbb{R}$. The same argument, compressing (3) to an eigenspace of L_0 , also proves that L_0 has simple spectrum.

3 Newton equations and global completeness

Choose a smooth unitary eigenbasis $U(t)$ of $X(t)$ and gauge its columns so that $e^*U(t) = (1, \dots, 1)$. The gauge is possible: the diagonal compression of (3) says that every overlap has modulus one. Put $\tilde{L} = U^*L_0U$. The off-diagonal part of the commutator gives

$$(x_j - x_k)\tilde{L}_{jk} = ig \quad (j \neq k). \quad (5)$$

Hermitian eigenvalue perturbation gives

$$\dot{x}_j = \tilde{L}_{jj}, \quad \ddot{x}_j = 2 \sum_{k \neq j} \frac{|\tilde{L}_{kj}|^2}{x_j - x_k} = 2g^2 \sum_{k \neq j} (x_j - x_k)^{-3}.$$

At $t = 0$, $(x_j, \dot{x}_j) = (q_j, p_j)$, so these are precisely Hamilton's equations for (1). Local uniqueness on the chamber identifies the pencil curve with the physical solution. Since $X(t)$ exists and remains simple for every real t , the solution is collision-free and complete in both time directions.

Finally $L(t) = U(t)^*L_0U(t)$, hence for every integer $m \geq 1$,

$$\text{Tr } L(t)^m = \text{Tr } L_0^m.$$

In particular (4) is the Hamiltonian integral.

4 A global spectral atlas

Let $\lambda_1 < \dots < \lambda_N$ be the simple eigenvalues of L_0 , and let v_a be normalized eigenvectors. Compressing (3) to $\mathbb{C}v_a$ shows $|e^*v_a| = 1$; fix the remaining phase by $e^*v_a = 1$. Define the real intercept coordinates

$$a_a = v_a^*Q_0v_a.$$

In this gauged L_0 -eigenbasis, the commutator determines every entry of $\tilde{Q}$:

$$\tilde{L} = \text{diag}(\lambda_1, \dots, \lambda_N), \quad \tilde{Q}_{aa} = a_a, \quad \tilde{Q}_{ab} = \frac{ig}{\lambda_b - \lambda_a} \quad (a \neq b). \quad (6)$$

Thus a phase point determines a unique pair $(\lambda_1 < \dots < \lambda_N, a \in \mathbb{R}^N)$.

Conversely, start with any such pair and define the Hermitian matrix $\tilde{Q}$ by (6). It obeys $[\tilde{Q}, \tilde{L}] = ig(J - I)$. The same compression/rank argument proves that $\tilde{Q}$ has simple spectrum. Order its eigenvalues and diagonalize it. The diagonal compression of the commutator makes every component of the transformed e have modulus one; a diagonal phase gauge makes all components equal to one. In that basis

$$Q = \text{diag}(q_1 < \dots < q_N), \quad L_{jk} = p_j \delta_{jk} + \frac{ig(1 - \delta_{jk})}{q_j - q_k},$$

with real p_j . This construction is inverse to the forward one, so (λ, a) is a global bijective spectral atlas, not merely asymptotic notation. No symplectic-form claim is needed here.

The finite certificate accompanying this note checks the Hermitian and commutator entries exactly on rational data, then reconstructs (6) numerically; it is regression evidence, not the proof of the all- N statement.

Round-1 boundary. This revision adds the global forward and inverse spectral atlas to the round-0 simplicity, Newton, completeness, and integral package. The two asymptotic ends, ordering reversal, aperiodicity, and the Route-A verdict are deferred to the final source revision.

Declarations

Data and code availability. Package-local deterministic code and exact evidence accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

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AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

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